

Identifying Markers of Immersion Using Auditory Event-Related EEG Potentials in a VR Jigsaw Puzzle

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ABSTRACT

Recent studies investigating the role of challenge immersion caused by task demand have focused on correlating auditory event-related potentials (ERPs) to difficulty. However, traditional approaches focus on the mean amplitude of the ERPs and may be compromised by confounding variables. This study introduces a novel experimental protocol for studying challenge immersion when playing a VR jigsaw puzzle overcoming the common confounding variables by analyzing the area under the ERP curve and peak-to-peak amplitude. Although preliminary, our work shows promise for auditory ERPs as immediate markers of immersion that can be used to increase the immersion-related performance and learning transfer in VR trainings.

Index Terms: Human-centered Computing—Human-computer Interaction (HCI)—Interaction Paradigms—Virtual Reality

1 INTRODUCTION

In virtual reality (VR), immersion is defined as a psychological state in which a person attends to the intended task and as a result does not pay attention to unrelated sensory stimuli inside or outside the virtual environment [1]. Immersion caused by the task demand is called challenge immersion, and immersion from the sensory information in the environment is called sensory immersion [2]. While subjective measures of quantifying challenge immersion based on self-reported questionnaires rely on participant understanding of the concept of being immersed, and their interpretation of questions, objective and quantifiable metrics have the advantage that they do not require conscious deliberation. One such method is high-end electroencephalography (EEG), which has been used to quantify immersion via measurement of the amplitude of event-related potentials (ERPs) to a stimulus which is not related to the VR task [2].

ERPs represent an average of the variations in the EEG signal of a person in response to a certain stimulus. The amplitude of ERP responses to task-irrelevant stimuli decreases when the attentional demands of the intended task increases. In other words, the amplitude of ERPs is dampened by the central nervous system, when attention is diverted to the intended task [3]. Recent studies typically use the mean amplitude of certain regions of the ERPs as measures of challenge immersion using the oddball paradigm with auditory stimulation, in which a modified auditory tone is played with a certain probability among other “standard” tones [4]. Such studies are lengthy to enable a sufficient number of oddball stimuli to create a robust average, which can result in confounding factors including, but not limited to fatigue and inattention that can interfere with the results [5]. Additionally, this study shows that the use of mean amplitude does not consider the cancellation of positive and negative ERPs for certain components and relies on using the slow wave (SW) components which increases the chances of their corresponding ERPs being compromised by the confounding variables.

To avoid the problems associated with the mean ERP amplitude and previous ERP-based protocols, our study investigates challenge immersion by manipulating task difficulty and measuring the amplitude of ERPs elicited by one type of auditory stimuli which is presented at random interstimulus intervals (ISIs), so that the brain

is not able to anticipate their arrival. In addition to the metrics established in the literature, two additional measurements of area under the curve and peak-to-peak amplitude are also included to overcome the challenges associated with mean amplitude measurements. We adapted a VR jigsaw puzzle game to accommodate for confounding variables, where we have equalized the image type (e.g., landscapes with similar elements). This approach overcomes a limitation associated with differences in difficulty introduced by having entirely different scenarios that may add stress or higher cognitive load, making it difficult to separate EEG differences due to immersion from other confounding variables [6]. We hypothesize that the evoked potential metrics would be lower in the harder condition than the easy condition.

2 METHODS

Two proof of principle protocols were used to investigate the feasibility of the chosen task for discriminating between low and high levels of immersion. The participants were required to sit on a chair while the Waveguard™ 64-electrode EEG cap and Oculus Quest ProVR headset were placed on their head. In both protocols, the participants started with a “baseline” data collection, in which they watch a 360° VR video of the study room for two 6-min blocks. The choice of showing a 360 video of the study room to ensure participants remain still and relaxed within a non-challenging scenario that is familiar to them. The calibration was followed by 4 6-min blocks of playing the jigsaw puzzle for which the order of the blocks is either easy-easy-hard-hard/hard-hard-easy-easy for the first protocol and easy-hard-hard-easy for the second protocol. The easy and hard tasks were differentiated by 40- and 72-piece puzzles in the first, and 24 and 60-piece puzzles in the second protocol. To have a benchmark for correlating the markers of difficulty with immersion, the second protocol also included having participants complete the 19-item presence questionnaire after completing the tasks [7]. During all six blocks, an auditory sine waveform (79 dB tones with 1KHz frequency) was played through two speakers located 50 cm behind the participant's head with a random ISI of 6–8 seconds (via LabVIEW code) to avoid anticipation by the participant. A TMSi REFA-8 amplifier was used for EEG recording. The length of the blocks and the ISI were chosen after careful consideration of the results of several pilot rounds conducted to ensure the best signal acquisition. As a proof of principle, four individuals (3 male, mean age 20) participated in the first protocol, and six individuals (2 male, mean age 21) participated in the second protocol.

ERP data was processed using Python and Google Collaboratory. The average of ERPs for each individual and the grand average for all participants had the baseline corrected using the 300 ms pre-stimulus period. The average DC for the first 300 ms was found and removed from the signal. Previous measures introduced for quantifying immersion through auditory ERPs include mean amplitudes for N1 (100–200 ms), SW1 (400–650 ms) and SW2 (650–900 ms) components ([4]), peak-to-peak amplitude for the highest negative and positive amplitudes (i.e. the N1-P2 complex) ([5]) and peak negative amplitude (N1). For this study, we included two additional metrics: 1- Area under the curve (AUC): the summation of integration of absolute value of the ERPs for the

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same N1, SW1 and SW2 components; 2- AUC for the N1-P2 complex: the summation of integration of absolute value of ERPs for 50-250 ms as shown in Figure 1.

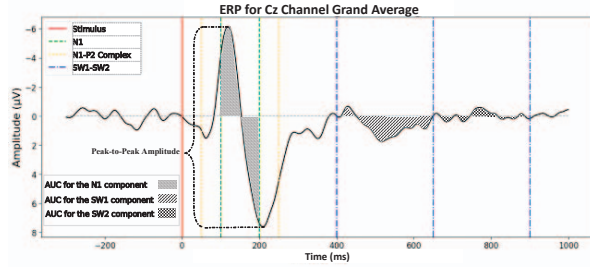


Figure 1: Graphical representation of the auditory ERP-based metrics for immersion.

3 RESULTS AND DISCUSSION

The results for both levels of difficulty are presented in Figure 2, including the mean amplitude of the ERPs for the N1-P2 and SW1 components (Figures 2a and 2b respectively), AUC for the N1 and SW1 and N1-P2 components (Fig. 2c, 2d and 2e respectively) and the mean amplitude for N1 components (Figure 2f). Table 1 presents the components collected from the frontal, parietal and central channels (i.e., Fz, Pz, and Cz, respectively), where the ERP amplitude was low for the task with the higher difficulty. For example, the mean amplitude for the difficult puzzle showed a significant difference in low latency component (N1) when compared to the easy one on the central and frontal channels in the second protocol (Figure 2a).

This is likely due to excitability difference in the neural generators of the N1 component. The primary sources of the N1 neural generators are situated within the superior temporal lobe, encompassing both the primary and secondary auditory cortices, as well as auditory association region [5]. The same significant difference is shown for AUC for the N1 for the same frontal and central channels (Figure 2c). Due to the immunity of the AUC to the cancellation of negative and positive components, the difference between channels is greater than for the mean amplitude. The mean total score for the PQ questionnaire for the easy condition was 117.13 (sd=11.16) and for the hard condition was 118.00 (sd=10.64). In the literature, a higher difficulty is generally associated with a higher sense of being immersed in the activity [2, 4]. However, the PQ questionnaire includes a number of questions unrelated to immersion, and as such may not be able to sensitively demonstrate the differences in immersion related to the two puzzle versions. Indeed, there is no pure "immersion" questionnaire, as the questionnaires are designed to measure the complexity of presence-related factors including immersion [8]. These results may suggest that ERP measures might be a more sensitive "moment by moment" measure of immersion.

Table 1: EEG Channels for which the mentioned metric is observed as expected.

Metric	Component	Channels	
		Protocol 1	Protocol 2
Mean Amplitude	N1(100-200 ms)	F3	F3, Fz, C3, Cz, P3,
	SW1(400-650 ms)	-	F3, C3, Pz
AUC	N1 (100-200 ms)	F3	F3, Fz, C3, Cz
	SW1 (400-650 ms)	-	C3, Pz
Peak-to-Peak	N1-P2(50-250 ms)	F3, P3	Pz, C3
AUC	N1-P2 (50-250 ms)	C3, Cz	Fz, P3

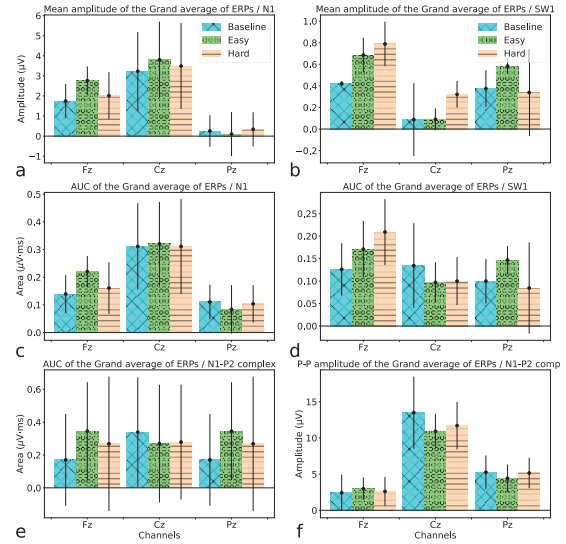


Figure 2: Proposed and established metrics for the 2nd protocol.

4 CONCLUSION

This study introduces a novel protocol using auditory ERP-based metrics to assess immersion by manipulating task difficulty. By presenting a task that is similar in all aspects except difficulty confounding variables such as fatigue, mood changes, boredom, etc. were minimized, thus enhancing the likelihood that observed ERP differences represented immersion differences. The AUC aligns with the widely used mean amplitude. Using targeted EEG electrodes in the proposed protocol, ERPs exhibit potential for distinguishing between difficulty levels. This "proof of principle" study with a small sample size, shows promise, suggesting avenues for future research. Larger studies and comparisons with existing metrics are essential to further validate and advance this work.

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